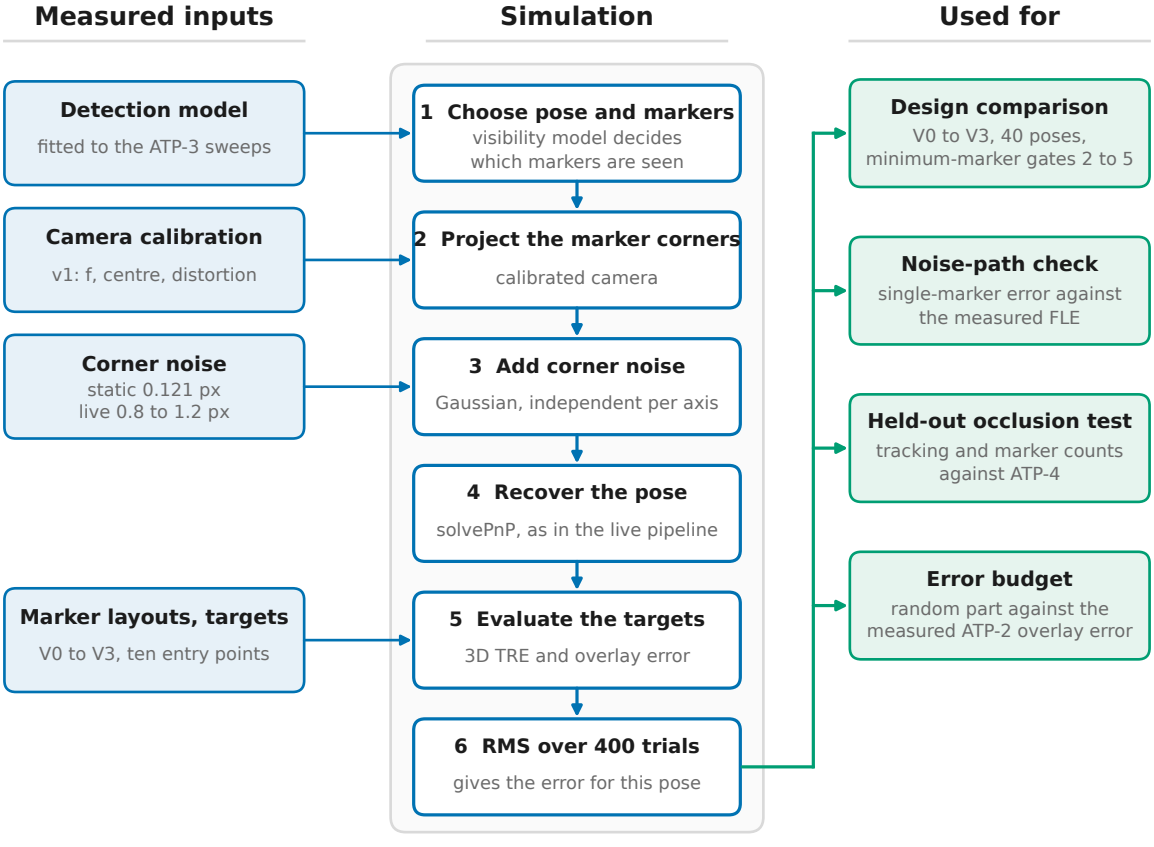
Measured inputs
Simulation
Used for
Detection model
fitted to the ATP-3 sweeps
Camera calibration
v1: f, centre, distortion
Corner noise
static 0.121 px
live 0.8 to 1.2 px
Marker layouts, targets
V0 to V3, ten entry points
1 Choose pose and markers
visibility model decides which markers are seen
2 Project the marker corners
calibrated camera
3 Add corner noise
Gaussian, independent per axis
4 Recover the pose
solvePnP, as in the live pipeline
5 Evaluate the targets
3D TRE and overlay error
6 RMS over 400 trials
gives the error for this pose
Design comparison
V0 to V3, 40 poses, minimum-marker gates 2 to 5
Noise-path check
single-marker error against the measured FLE
Held-out occlusion test
tracking and marker counts against ATP-4
Error budget
random part against the measured ATP-2 overlay error